

MYAN GUPTA

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SUMMARY

Computer Science student and software engineer focused on backend systems, distributed systems, and AI/ML infrastructure, with research experience building evaluation workflows for multi-agent Large Language Model systems. Built Go-based distributed storage, FastAPI/React forecasting platforms, Reinforcement Learning benchmarks, and automation pipelines using Docker, AWS, REST APIs, and Python. Strong foundation in operating systems, concurrency, databases, algorithms, and systems programming, with a focus on measurable latency, reliability, and scalability improvements.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, C, JavaScript, TypeScript, SQL, Go, OCaml
- **Backend & Cloud:** FastAPI, Flask, REST APIs, JSON-RPC, Docker, Kubernetes, AWS EC2, AWS S3, AWS Lambda, CI/CD, Linux/Unix
- **AI/ML & Data:** PyTorch, Hugging Face, OpenAI API, MLX, Pandas, NumPy, SARIMA, Prophet, Federated Learning
- **Tools & Core:** Git, Playwright, Redis, PostGIS, Pydantic, GDB, Valgrind, ThreadSanitizer, k6, Load Testing, Concurrency

EDUCATION

University of Minnesota

May 2027

B.S., Computer Science

- **GPA:** 3.8
- **Achievements:** Dean's List, Google AI Agents Intensive
- **Coursework:** Data Structures and Algorithms, Operating Systems, Software Engineering, Databases, Artificial Intelligence, Machine Architecture, Modern Cryptography, Statistics, Programming Languages, Computer Networks, Linear Algebra, Discrete Mathematics, Computer Graphics, Systems Programming

EXPERIENCE

University of Minnesota

Research Assistant

Jan 2025 - Present

- Built trace-analysis workflows for multi-agent LLM systems using Python and FastAPI, categorizing system design, inter-agent coordination, and task-verification failures across execution runs, which helped the team pinpoint design flaws more quickly
- Developed scalable evaluation pipelines inspired by MAST-Data to analyze 1,600+ annotated traces across 7 agent frameworks and 14 failure modes, improving root-cause diagnosis for unreliable agent behavior.
- Implemented LLM-as-a-judge style validation and reporting scripts tracking accuracy, agreement, latency, throughput, and failure distribution, supporting reproducible model and system comparisons.

Teaching Assistant

Sep 2024 - Present

- Mentored 100+ students in Java, Python, data structures, algorithms, and systems programming, helping improve exam performance by 20%.
- Designed labs for graph algorithms, recursion, and dynamic programming, increasing average lab scores by 25% through targeted practice and debugging support.
- Diagnosed Linux memory, runtime, and performance issues using GDB and Valgrind, improving student code reliability and submission quality.

PROJECTS

Reinforcement Learning Benchmarking Platform

Jan 2026 - Present

- Built an OpenEnv-compatible Reinforcement Learning evaluation backend for email workflows, achieving 0.967 mean return across benchmark tiers.
- Developed an async JSON-RPC MCP server with Pydantic validation to run 20+ classification, drafting, and workflow scenarios through a standardized API.
- Added deterministic grading, trace logging, and integration tests across 3 benchmark tiers, improving reproducibility for multi-step agent tasks.

ICU Capacity Forecasting Platform

Feb 2026 - Mar 2026

- Deployed a containerized ML forecasting platform on AWS EC2, S3, and Lambda with REST APIs and a React dashboard to support hospital capacity planning.
- Optimized API queries and concurrency bottlenecks under load, reducing P95 response latency from 360ms to 14ms for forecast retrieval.
- Integrated SARIMA, Prophet, and federated learning across 8 hospital clients, improving model robustness on non-IID healthcare time-series data.

Automated Job Application Pipeline

Mar 2026 - Apr 2026

- Engineered browser automation across Greenhouse, Lever, Workday, and other ATS flows, scaling repetitive job submissions to 200+ tracked workflows.
- Fine-tuned Phi-3.5 Mini with 4-bit MLX QLoRA on Apple Silicon, replacing external API calls for local inference and reducing generation cost to zero.
- Added JSONL logging, structured tracking, and retry-safe execution, reducing manual submission time by 90% while improving auditability.

Consistent Hashing Key-Value Store

Apr 2026 - Present

- Built a distributed key-value store in Go with consistent hashing to route keys across multiple backend nodes, improving horizontal scalability and minimizing key remapping during node changes.
- Containerized router and storage nodes with Docker and exposed REST APIs for health checks, key-value operations, ring inspection, and node debugging to support reliable local cluster testing.
- Benchmarked the system using k6 load tests with 20 virtual users and 60K operations, validating sub-500ms P95 latency and low request failure rate under concurrent workloads.